

Stress & deflection Equations

2 basic eqns. for design of helical springs

Load-stress eqn

Load-deflection eqn.

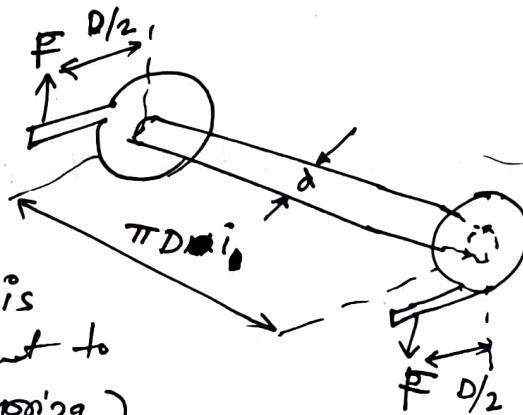
$D \rightarrow$ mean coil dia

$d \rightarrow$ wire dia.

No. of active coils in spring = i .

Spring subjected to axial force = F .

(When wire of helical spring is uncoiled & straightened, it takes shape of a bar.)



(While deriving bar is considered equivalent to actual helical spring).

Dia of bar = wire dia of spring = d .

Length of equivalent bar = length of one coil in spring (πD) $\times$ no. of active coils (i)
 $= \pi D i$

Bar is fitted with bracket @ both end

Length of bracket = mean coil radius of spring = $D/2$.

Force, F acting @ end of bracket

↓
Induces torsional shear stress in bar.

$$\text{Torsional moment } M_t = F \times \frac{D}{2}$$

Torsional shear stress in a bar is given by.

$$\tau_t = \frac{16 M_t}{\pi d^3}$$
$$= \frac{16 \times F \times \frac{D}{2}}{\pi d^3}$$

$$\boxed{\tau_t = \frac{8FD}{\pi d^3}}$$

$$\frac{D}{d} = C$$

When equivalent bar $\rightarrow$ bent into helical coil

↓
additional stresses occur

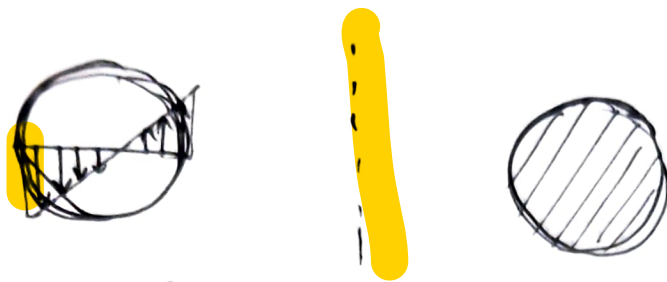
Direct / transverse shear stress in spring wire.

~~Torsional shear~~
(already considered)

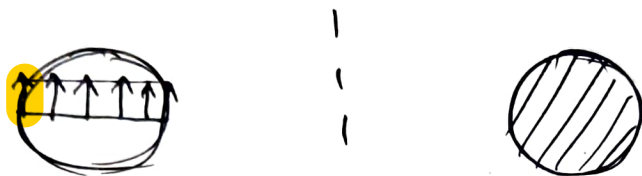
Length of inside fibre is less than length of outside fibre

↓
this causes stress concentration @ inside fibre of coil.

↓
Curvature effect



(pure torsional stress)



(direct shear stress)



(combined torsional, direct & curvature shear stresses)

Egn. should be modified to consider all the stresses,
So, Let's assume 2 factors for account of these effects.

K_d → factor to account for direct shear stress.

K_c → factor to account for stress corr. due to curvature effect.

So, combined effect of these 2 factors given by,

Stress correction factor → $K = K_d \cdot K_c$

Direct stress (for the bar) is given by eqn,

$$\tau_d = \frac{F}{\frac{\pi d^2}{4}} = \frac{4F}{\pi d^2} = \frac{8FD}{\pi d^3} \left(\frac{0.5d}{D} \right)$$

So, the resultant shear stress is spring wire,

$$\tau = \tau_t + \tau_d$$

$$= \frac{8FD}{\pi d^3} + \frac{8FD}{\pi d^3} \left(\frac{0.5d}{D} \right)$$

$$\tau = \frac{8FD}{\pi d^3} \left[1 + \frac{0.5d}{D} \right]$$

shear stress correction factor,

$$K = 1 + \frac{0.5d}{D}$$

$$K = 1 + \frac{0.5}{C}$$

$C \rightarrow$ Spring Index.

$$C = \frac{D}{d}$$

$K \rightarrow$ stress factor ~~Wahl factor~~.

$$\tau = K \cdot \left(\frac{8FD}{\pi d^3} \right)$$

This is eqn. for resultant stress which includes torsional shear stress, direct shear stress & stress conc. due to curvature.

$K \rightarrow$ Wahl factor / stress factor

$$K = \frac{4C-1}{4C-4} + \frac{0.615}{C}$$

Angle of twist (θ) for equivalent bar

$$\theta = \frac{M_t \times l}{JG}$$

$$\frac{I}{J} = \frac{G\theta}{l}$$

$M_t \rightarrow$ torsional moment $= F \times D/2$

$l \rightarrow$ length of bar $= \pi D i$

$J \rightarrow$ polar moment of inertia of bar $= \pi d^4/32$

$G \rightarrow$ modulus of rigidity.

$$\therefore \theta = \frac{\frac{FD}{2} \times \pi D i}{\frac{\pi d^4}{32} \times G}$$

$$\theta = \frac{16 F D^2 i}{G d^4}$$

Axial deflection y of spring for small values of θ ,

$$y = \theta \times \text{length of bracket}$$
$$= \theta \times D/2$$

$$\therefore y = \frac{8 F D^2 i}{G d^4} \times \frac{D}{2}$$

$$y = \frac{8 F D^3 i}{G d^4}$$



SPRINGS

Helical Spring

Design procedure for Helical Spring

Step-1

Design of wire diameter 'd'

$$\tau = \frac{8FDK}{\pi d^3} \quad [\text{page 169, Eqn 11.1 (d)}]$$

$$c = D/d$$

$$\therefore \tau = \frac{8FcK}{\pi d^2}$$

$c \rightarrow$ spring index
 $K \rightarrow$ stress correction factor

$$K = \frac{4c-1}{4c-4} + \frac{0.615}{c} \quad (\text{stress } \overset{\text{Correction}}{\cancel{\text{correction}}} \text{ factor})$$

[page 169, Eqn 11.2 (a)]

Step-2

Mean coil diameter, $D = cd$.

$$D_i = D - d, \quad D_o = D + d$$

$D_i \rightarrow$ Inside diameter of the coil.

$D_o \rightarrow$ Outside diameter of the coil.

Step-3

Find the deflection,

$$y = \frac{8FD^3i}{Gd^4} = \frac{8Fc^3i}{Gd}$$

[page 170, Eqn 11.5 (a)]

Step-4

Free length of the spring.

$$l_0 \geq (i+n)d + y + a \quad [\text{page 174, Eqn 11.18}]$$

$a \rightarrow$ clearance between the coil.

$$\underline{a \approx 0.28y} \quad 159$$

$a \rightarrow$ total clearance between working coils

Step-5

Pitch.

[Table 11.4, page 189]

Step-6

$$\text{Stiffness, } F_0 = \frac{F}{y} \quad [\text{page 173, Eqn 11.14(d)}]$$

Step-7

Length of the wire required to manufacture the spring.

$$l = (i_t \times \pi \times D)$$

$$\text{where, } i_t = (i+n)$$

$i_t \rightarrow$ total no. of turns.
Page 189 $\rightarrow$ select from table 11.4.

Design problem for helical spring

- ① Design a helical compression spring to sustain an axial load of 3 kN. The deflection is 60 mm. and the spring index is 6. The shear stress is not to exceed 300 MPa. The rigidity modulus for spring material is 81 GPa.

Given data

$$\text{Axial load, } F = 3 \text{ kN} = 3 \times 10^3 \text{ N}$$

$$\text{Deflection, } y = 60 \text{ mm.}$$

$$\text{Spring index, } C = 6 = \frac{D}{d}$$

$$\tau = 300 \text{ MPa}$$

$$G = 81 \text{ GPa} = 81 \times 10^3 \text{ MPa}$$

Step-1

Design of wire diameter.

$$\tau = \frac{8Fck}{\pi d^3}$$

$$K = \frac{4C-1}{4C-4} + \frac{0.615}{C} = \underline{\underline{1.2525}}$$

$$\tau = \frac{8 \times 3 \times 10^3 \times 6 \times 1.25}{\pi d^3}$$

$$300 = \frac{8 \times 3 \times 10^3 \times 6 \times 1.25}{\pi d^3}$$

$$d = \underline{\underline{13.8 \text{ mm}}} \approx \underline{\underline{14 \text{ mm}}}$$

Step-2

$$\text{Mean coil diameter, } D = c \times d \\ = 6 \times 14 = \underline{\underline{84 \text{ mm}}}$$

$$\text{Inside dia. of coil, } D_i = D - d \\ = 84 - 14 \\ = \underline{\underline{70 \text{ mm}}}$$

$$\text{Outside dia of coil, } D_o = D + d \\ = 84 + 14 \\ = \underline{\underline{98 \text{ mm}}}$$

Step-3

$$\text{Deflection, } y = \frac{8 F c^3 i}{G d} = \frac{8 \times 3 \times 10^3 \times 6 \times i}{81 \times 10^3 \times 14}$$

$$60 = 4.57 i$$

$$i = 13.125 \approx \underline{\underline{14}}$$

Step-4

Free length of spring

$$l_0 \geq (i+n) \phi + y + a$$

$$a \approx 0.25 y.$$

Since end condition is not given,

Assume the end condition is square and grounded

So the value of $n=2$.

$$a = 0.25 \times \overset{60}{\cancel{14}} = \underline{\underline{15}}$$

$$l_0 \geq 224 + 60 + 15$$

$$l_0 \geq \underline{\underline{299 \text{ mm}}}$$

→ page 174, Eqn 11.18(a)

✓
Page 189 → choose 'n' from table - (assume condition if not given)

Step-5

Pitch

(Table 11.4, Page 189)

$$P = \frac{(L_0 - 2d)}{i} = \frac{(299 - 2 \times 14)}{14} = \underline{\underline{19.35 \text{ mm}}}$$

$$\therefore P \approx \underline{\underline{20 \text{ mm}}}$$

Step-6

$$\text{Stiffness, } F_0 = \frac{F}{y} = \frac{3 \times 10^3}{60} = \underline{\underline{50 \text{ N/mm}}}$$

Step-7

Length of the wire required to manufacture the spring.

$$L = (i_t \times \pi \times D)$$

$$\text{where, } i_t = (i + n)$$

$$i_t = 14 + 2 = \underline{\underline{16}}$$

$$L = 16 \times \pi \times 84$$

$$= \underline{\underline{4222.300 \text{ mm}}}$$

② Design a Helical compression spring for a spring loaded safety valve operating pressure 1 MPa. Maximum pressure when the valve blows off freely = 1.1 MPa. Maximum lift of the valve when the pressure is 1.1 MPa = 6 mm. Diameter of the valve = 100 mm. Maximum shear stress = 360 MPa. Modulus of rigidity = 84 GPa. Spring index = 5.5 ?

Note: In this problem, load & deflection (y) is not given.

Given

Initial pressure $P_1 = 1 \text{ MPa}$

Max. pressure $P_2 = 1.1 \text{ MPa}$.

$$G = 84 \text{ GPa} = 84 \times 10^3 \text{ MPa}.$$

$$C = 5.5 = \frac{D}{d}.$$

Lift, $y' = 6 \text{ mm}$.

Dia. of valve, $D_v = 100 \text{ mm}$.

$\tau = 360 \text{ MPa}$.

Area of the valve, $A_v = \frac{\pi}{4} \times D_v^2$
 $= 7853.98 \text{ mm}^2$

(min. load) $F_1 = P_1 \times A = 1 \times 7853.98 = 7853.98 \text{ N}$

(max. load) $F_2 = P_2 \times A = 1.1 \times 7853.98 = 8639.378 \text{ N}$

$$\boxed{\frac{F_1}{y_1} = \frac{F_2}{y_2} = \frac{F_2 - F_1}{y'}}$$

Page 170, Eqn 11.7(a)
11.7(b).

$$\frac{F_2 - F_1}{y'} = \frac{8639.378 - 7853.98}{6} = 130.89$$

$$\therefore \frac{F_2}{y_2} = 130.89 \Rightarrow y_2 = \frac{F_2}{130.89} = \frac{8639.378}{130.89} = 66 \text{ mm}$$

$$\frac{F_1}{y_1} = 130.8 \rightarrow y_1 = \frac{F_1}{130.8} \quad \therefore y_1 = \underline{\underline{60 \text{ mm}}}$$

We are designing the spring for maximum force and for maximum deflection.

$$\therefore \text{choose, } F = F_2 = \underline{\underline{8639.378 \text{ N}}}$$

~~(Stiffness)~~
(Load) and.

$$y = y_2 = \underline{\underline{66 \text{ mm}}}$$

(deflection)

Step-1

Design of wire diameter (d)

$$\tau = \frac{8Fck}{\pi d^3}$$

$$k = \frac{4C-1}{4C-4} + \frac{0.615}{C} = \underline{\underline{1.27}}$$

$$360 = \frac{8 \times 8639.378 \times 5.5 \times 1.27}{\pi \times d^3}$$

$$d = 20.6 \simeq \underline{\underline{21 \text{ mm}}}$$

Step-2

Mean coil diameter,

$$D = C \times d$$

$$= \underline{\underline{115.5 \text{ mm}}}$$

$$\text{Outside dia, } D_o = D + d = \underline{\underline{136.5 \text{ mm}}}$$

$$\text{Inside dia, } D_i = D - d = \underline{\underline{94.5 \text{ mm}}}$$

Step-3

$$\text{Deflection } y = \frac{8Fc^3i}{6d.}$$

$$66 = \frac{8 \times 8639.318 \times (5.5)^3 \times i}{84 \times 10^3 \times 21}$$

$$i = 10.12$$

(no. of active turns).

$\therefore$ Take $i = 11$ (Take next whole number)

Step-4

Free length,

$$l_0 \geq (i+n)d + y + a$$

$$\begin{aligned} a &= 0.25 \times y \\ &= 0.25 \times 66 \\ &= \underline{16.5 \text{ mm}} \end{aligned}$$

Assume end condition is Square & grounded.
So, the value of $n=2$.

$$l_0 \geq (11+2)21 + 66 + 16.5$$

$$l_0 \geq \underline{355.5 \text{ mm}}$$

$$l_0 \geq \underline{356 \text{ mm}}$$

Step-5

Pitch, Table 11.4, page 189.

Assume
Square & grounded

$$P = \frac{l_0 - 2d}{i} = \frac{356 - 2 \times 21}{11} = \underline{28.54 \text{ mm}}$$

$$\therefore P \approx \underline{29 \text{ mm}}$$

Step-6

$$\begin{aligned}\text{Stiffness, } F_0 &= \frac{F}{y} = \frac{8639.378}{66} \\ &= 130.89 \\ &\underline{\underline{\approx 131 \text{ N/mm}}}\end{aligned}$$

Step-7

Length of the wire required to manufacture the spring.

$$l = i_t \times \pi \times D$$

$$\begin{aligned}i_t &= (i + 2) \longrightarrow \text{page 189, Table 11.4.} \\ &= 11 + 2 \\ &= \underline{\underline{13}}\end{aligned}$$

$$\begin{aligned}\therefore l &= 13 \times \pi \times 15.5 \\ &= \underline{\underline{4717.10 \text{ mm}}}\end{aligned}$$